

QUALITY ANALYSIS OF BOREWELL WATER

Submitted in partial fulfilment of the requirements of
the course **Innovative Product Development-III**

Year 3, Sem V Computer Engineering

By

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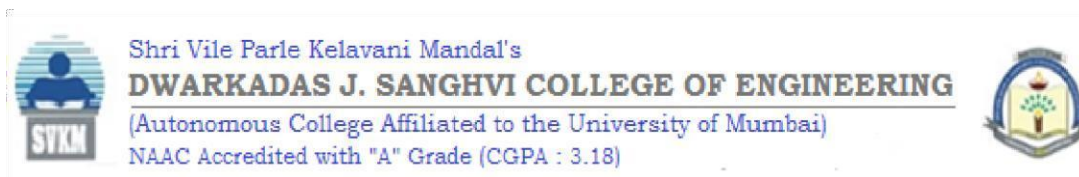
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2025 – 2026

CERTIFICATE

This is to certify that the project entitled **“Quality Analysis of Borewell Water”** is a Bonafide work of **“Ishaan Sheth”(60004230217)** ,**“Jainam Gala”(60004230107)** ,**“Kavya Shah”(60004230107)** , **“Parth Mehta”(60004230002)** submitted as a project work for the course **Innovative Product Development-III, Year 3, Semester V, TY B. Tech Computer Engineering**

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Examiners

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Contamination of borewell water is a pressing global issue due to increased levels of turbidity, ammonia, dissolved solids and chemical contaminants. This project outlines an innovative solution that will enable real-time monitoring of borewell water quality using an IoT based sensor network and ML models to forecast water safety metrics based upon quantitative data. The sensor array comprised several sensors including pH, turbidity, Electrical Conductivity (EC), DO and MQ135 gas sensors and an ESP32-S3 microcontroller for real-time, accurately calibrated monitoring of the physical and chemical parameters of the water sample. Streaming real-time data from the sensors enables instant alerts to be sent to the user via mobile and web dashboards, therefore allowing for user-friendly access and unlimited data storage. In addition, historical data from government sources on the long-term water quality trends was analysed in order to develop four different predictions based upon the historical data that was obtained: ARIMA, SARIMA, Prophet and Long Short-Term Memory (LSTM). The results indicate a wide degree of variance in the Water Quality Index (WQI) over the period from 2018 -2024, with a WQI range of between 110 - 126.5 observed over the entire study period which indicates continual degradation of groundwater quality. The performance of the various models was significantly different based on comparative analysis, illustrating differences between them. While ARIMA and SARIMA were able to identify the larger patterns of each model, their 95% confidence intervals indicated substantial uncertainty in the results (e.g. ARIMA: 114.92-151.62, SARIMA: 116.28-150.89 for Dec 2023). Prophet produced smoother estimates within its confidence interval range of 116.99-138.56. In comparison, the LSTM model was evaluated as the most accurate of the four using the lowest RMSE at 12.59, MAE at 10.41 and MAPE at 7.19%, and the narrowest prediction interval (5.40). Furthermore, LSTMs produced frequent true values that matched closely to those of WQI predictions within the range of 131-142, thus representing the most reliable and accurate method for determining WQI predictions in near real-time. By using ML-generated analytics combined with sensor-based monitoring to create this predictive tool, this new approach demonstrates improved time efficiencies, better precision and increased confidence levels when compared to traditional laboratory testing methods. The empirical data support this model's ability to serve as a predictive tool for early contamination detection, water resource planning, safety interventions at the community level, and other humanitarian uses.

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List of Abbreviations

Sr. No.	Abbreviation	Expanded form
1.	DSS	Decision Support System
2.	IoT	Internet of Things
3.	AI/ML	Artificial Intelligence/Machine Learning
4.	pH	Potential of Hydrogen
5.	TDS	Total Dissolved Solids
6.	GIS	Geographic Information System
7.	Turbidity	Measurement of Water Clarity

1. INTRODUCTION

Provision of potable water is essential for the promotion and maintenance of public health and agricultural growth, as water is a major input for industrial processes. In most areas, especially the rural and semi-urban areas, groundwater from bore wells is the chief means for supplying water to people [20]. Unfortunately, this water is more easily contaminated both naturally and through man-made sources, especially industrial effluents, agricultural runoff, and natural geological factors. Conventional methods used for determining water quality, which include laboratory testing and geophysical exploration, are mostly known for their low efficiency, which is marked by their sluggishness, high cost, and negative environmental impacts [11].

In overcoming the mentioned issues, the present study proposes a novel, real-time water quality monitoring system for borewell water using Internet of Things (IoT) technology, smart sensors, and Artificial Intelligence/Machine Learning (AI/ML) algorithms [7]. This system is used to monitor important water quality parameters like pH, turbidity, dissolved oxygen, TDS, and ammonia in a real-time manner. With the implementation of AI/ML models, it is possible to forecast trends of contamination and detect anomalies, thus enabling stakeholders to take preventive or remedial action in a timely manner.

The solution is cost-effective, scalable, energy-efficient, and sustainable. It also supports international efforts such as the United Nations Sustainable Development Goal 6 (Clean Water and Sanitation) that seeks to ensure the availability of safe water for all. The paper describes the motivation behind the product, describes the limitations of existing systems, defines the problem, describes the goals and applications of the product, and describes the proposed design and implementation strategy, including the application of the Water Quality Index (WQI) for assessing water safety.

2. NEED OF THE PRODUCT

2.1 Requirement of the product.

The need for an advanced borewell water quality monitoring system rose quickly the dependence increasing on groundwater as the primary source of water for millions of households, agricultural activities, industrial processes and much more. Groundwater contamination has been steadily increasing due to factors such as agricultural runoff containing fertilizers and pesticides, untreated industrial waste, seepage from septic tanks and natural mineral deposits. These pollutants adversely change chemical characteristics of water, such as turbidity, pH, dissolved oxygen, total dissolved solids, ammonia concentration, and hardness which makes the water unsafe for consumption and utility [1]. Even after most contamination remains undetected because water is often assumed to be clean simply due to its underground origin.

Traditional water testing processes require manual sample collection and laboratory analysis, which are expensive, time consuming, and impractical for frequent monitoring. These results usually take several days to be returned, by which contamination caused by time may worsen, and corrective actions won't be performed on a timely basis. Communities with limited financial resources or lack of awareness rarely test their borewell water, resulting in long term use of unsafe water. The consequences include chronic health disorders such as kidney failure, neurological diseases, gastrointestinal infections, and growth issues caused by continued exposure to harmful chemical constituents.

A real-time water monitoring system with the aid of IoT and machine learning is very essential for tackling these issues [14]. Through the systematic observation of water and the immediate notifications sent when the water crosses the safe limit, the system helps in making rapid decisions for the execution of preventive measures to avoid the dangers of diseases and failures. Thanks to the additional support of machine learning, there will be forecasted future water quality patterns and the systematic observation of contamination patterns for planning future courses of action instead of responding to adverse situations. Though the dependency on borewell water and the shortcomings of the present water monitoring systems are vast, the need for the proposed device is very essential to ensure the safe and accessible availability of water, which is the most essential commodity in life.

2.2 Drawbacks of the existing product

The traditional water analysis systems rely completely on conventional laboratory analysis, periodic water sampling, and simple portable devices. Although these systems have been around for several decades and offer a good level of accuracy, there are some crucial drawbacks that make them unfeasible and unreliable for actual practice. Laboratory analysis needs the physical transport of the water samples, which requires technical expertise and a substantial amount of time to produce the results. The lack of immediacy for actual effective measures during a contamination event. The periodic nature of the samples allows the results to offer a mere snapshot of the situation instead of analyzing the constituents of the water that follow daily and seasonal changes [8].

Although technological alternatives like the use of portable sensors are available, they are not integrating, storing data in a remotely accessible manner, identifying unusual data, or analyzing data for prediction. Text alerts are not provided, and these technologies are not designed for mobile and cloud platforms and provide negligible help in decision-making through trending. Text alerts are not given, and these technologies are not designed for mobile and cloud platforms and provide negligible help in decision-making through trending. In general, these technologies are not providing data transparency to the local population, and data is not made available in a local and timely manner. Data from these technologies is not providing an efficient way for local administrations and/or local organizations like the environment department to control water resources and develop preventive measures for their protection [8].

Another major disadvantage is that there is no predictive forecasting at all. Traditional approaches can characterize only the prevailing state of water quality and cannot predict further deterioration in the future, nor can they find the causes for such contamination patterns. Thus, responses continue to be reactive rather than proactive. In the face of increasing environmental degradation and increasing dependence on groundwater [10], existing approaches no longer suffice to guarantee safety, sustainability, and accountability. The proposed IoT-ML based monitoring system overcomes these deficiencies by enabling continuous real-time data collection, smart automated alerts, wireless accessibility, predictive machine learning analysis, and user friendly interfaces. This transition from manual, delayed, labour-intensive processes to automated intelligent evaluation makes the new system a necessary and transformative extension of previous solutions.

2.3 Applications of the Product

The bore well water quality monitoring system has a multitude of applications, including all types of household settings, agriculture units, industries, government water resource management units, and health organizations. In the case of households, the water quality monitoring system will help households check the quality of the water they utilize daily. With the instant update provided through a mobile or web facility, the water quality monitors will be less exposed to water-borne diseases[6]. In the case of agricultural units, water quality is a critical aspect that determines the productivity of the crop, fertility of the soil, and nutrition of the food produced. Water that is contaminated with excess salt, ammonia, or heavy metals can result in the destruction of the fertilizer composition of the soil. It can also affect plant growth. The instant update facility will help farmers identify the source of water to be used for irrigation, optimize fertilizer composition, and safeguard crop production.

Industrial estates and factories often require borewell water for various processes, such as cooling, washing, and production cycles. Poor quality groundwater may also increase the cost of maintenance of machinery in addition to damaging equipment and breaking operational processes. Automated monitoring does ensure that the regulatory standards are met without attracting penalties while allowing profitability in production. Government departments with the mandate to promote public health, urban development, and environmental safety can tap into data generated by the pervasive deployment of such sensors to make strategic water management decisions, react to emergencies involving water contamination, and make predictions about possible future water risks using predictive analytics [19]. Intelligent monitoring of this kind creates a data-driven approach that eventually leads to national vision contributions to health areas such as Sustainable Development Goal 6 - Clean Water and Sanitation.

In addition to such immediate economic and health spinoffs, this system enhances community consciousness and a sense of accountability on the part of citizens to have right information about water resources that they utilize. This is a society that is better informed and educated to make optimal use of resources. Essentially, this particular system is more than a mere innovation; it is a solution that sustains a very essential natural element, secures public health, and is crucial for boosting agricultural and industrial production.

3. SURVEY CONDUCTED

3.1 Field Survey

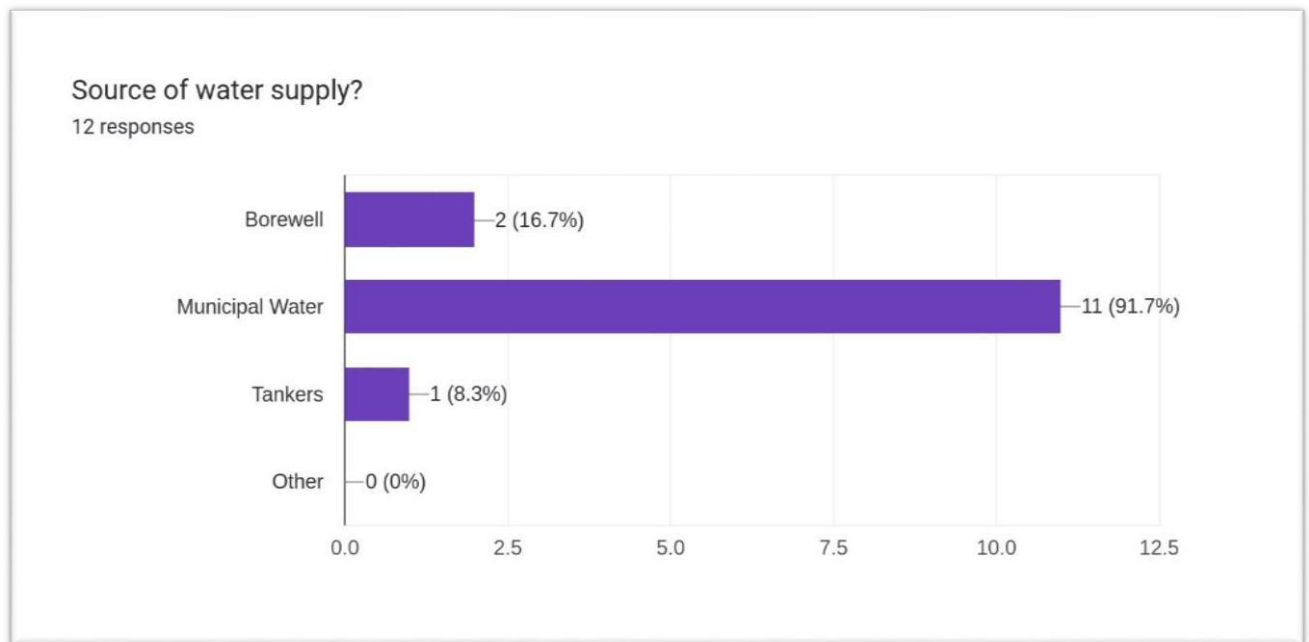


Fig 3.1.1 : Source of Water Supply

This bar chart, as shown in Fig. 3.1.1, displays the proportion of the water sources used by 12 participants. The majority of them (91.7%) use municipal water, followed by a lesser proportion using borewell water (16.7%) and tankers (8.3%). None of the participants used other sources of water. Inference:

Heavy dependence on municipal water suggests that public water facilities are highly developed and accessible to most of the respondents. But the prevalence of borewell and tanker use—though limited—may suggest supplementary usage or instances of poor or insufficient municipal supply in certain places.

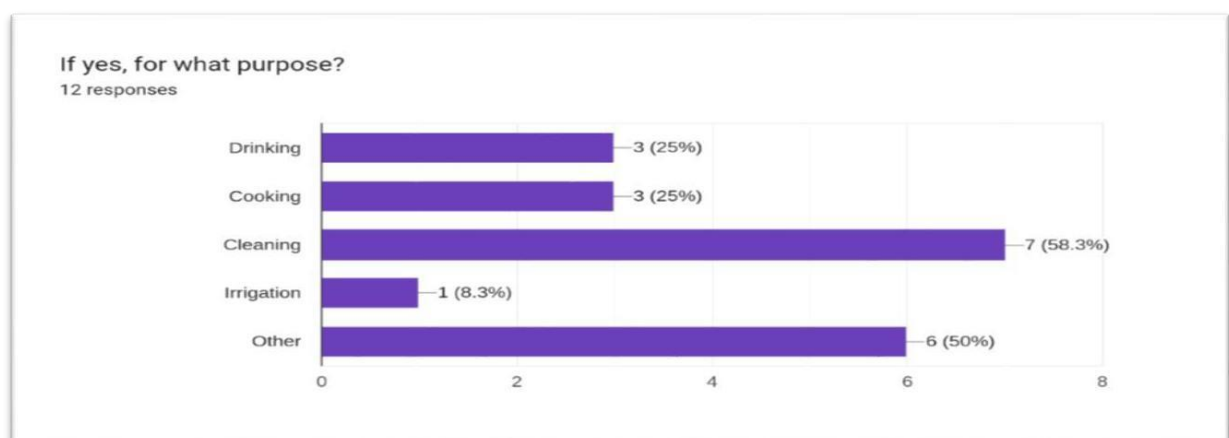


Fig 3.1.2 : Usage of Water Supply.

The bar chart, as shown in Fig. 3.1.2, displays the purpose for which the water is being used, from the responses of 12 individuals. Cleaning is the most common purpose mentioned (58.3%), followed by others (50%), and drinking and cooking each account for 25% of the responses.

Irrigation use is the least mentioned (8.3%).

Inference:

The data reveal water is primarily utilized for non-consumptive uses such as cleaning, which indicates a high level of daily water utilization going into sanitation and hygiene. The relatively lower utilization for drinking and cooking can reveal the existence of alternative water sources for drinking, or utilization of filtered/bottled water for drinking and cooking. The low utilization of irrigation indicates minimal agricultural activities among the respondents.

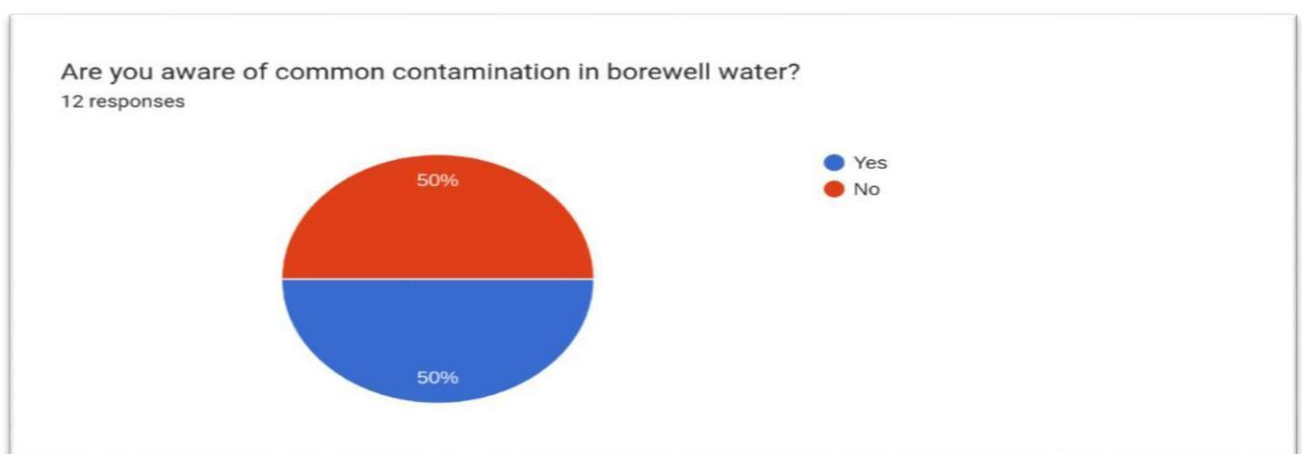


Fig 3.1.3 : Common contaminations in Borewell Water

The pie chart, as evident from Fig. 3.1.3 below, informs us on the level of awareness among the respondents with respect to the general aspects of contamination in borewell water. Half or 50% out of the 12 respondents were aware of the water contamination aspects. The other 50% were not aware.

Inference:

The results indicate a dichotomy of public opinions regarding contamination of borewell water. This indicates the necessity for greater public education and awareness of water quality and potential health risks of untreated or inadequately treated borewell sources. Public awareness can reduce health risks and promote improved water management.

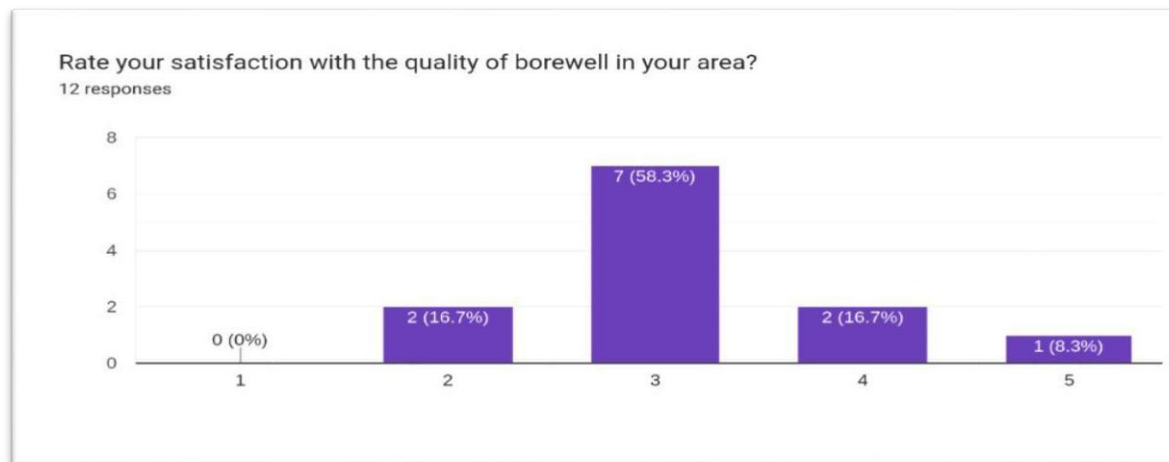


Fig 3.1.4 : Satisfaction Data of Borewell Water in the area

This bar chart in Fig. 3.1.4 indicates the level of satisfaction (score of 1 to 5) of respondents with the quality of borewell water in the locality. Most of them (58.3%) were neutral with a score of 3, and 16.7% of them were satisfied (score of 2) or dissatisfied (score of 4) each. A mere one respondent (8.3%) provided a high score of 5, and nobody provided even the lowest score of 1.

Inference:

Dominance by neutral ratings shows that while the borewell water quality is not poor, it is not as good as for most of the residents. Low percentage of high satisfaction ratings suggests that quality enhancement or treatment would be necessary. Lack of very low ratings (1) might suggest that critical issues do not occur very often, but total water quality is an issue or unknown variable.

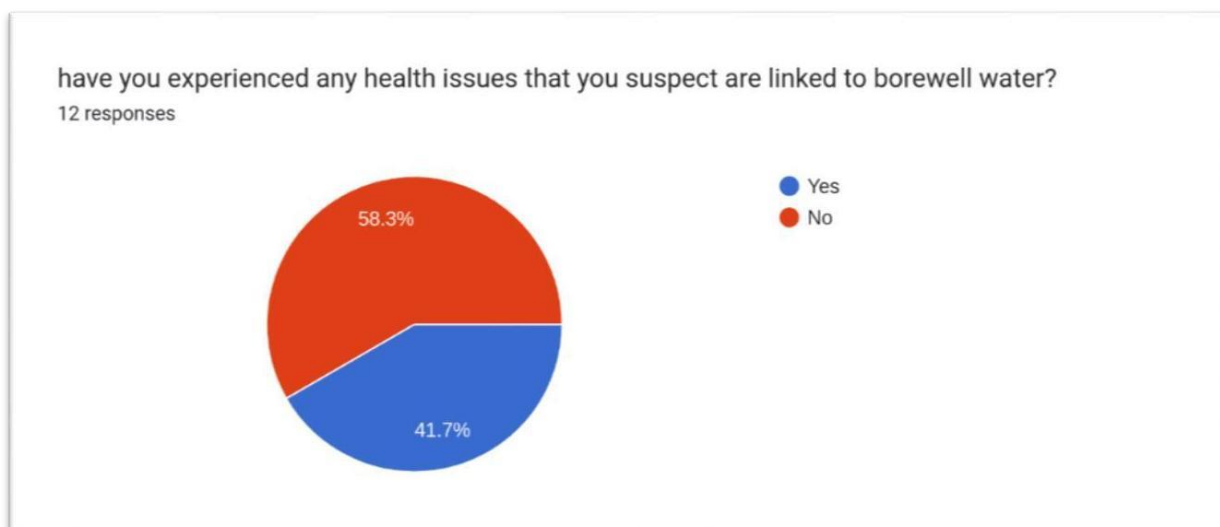


Fig 3.1.5 : Health issues which are linked to borewell water

This pie chart in Fig.3.1.5 represents the percentage of the interviewees who opined that they were affected by health issues as a result of borewell water. 41.7% out of 12 respondents said "Yes" while 58.3% said "No".

Inference:

A substantial proportion of the surveyed (nearly 42%) attribute their health problems to the use of borewell water, which may be an indicator of potential contamination or quality issues. Although none of them reported any such problem, the large minority reinforces the need for regular testing and education regarding the safety of borewell water to prevent potential health issues.

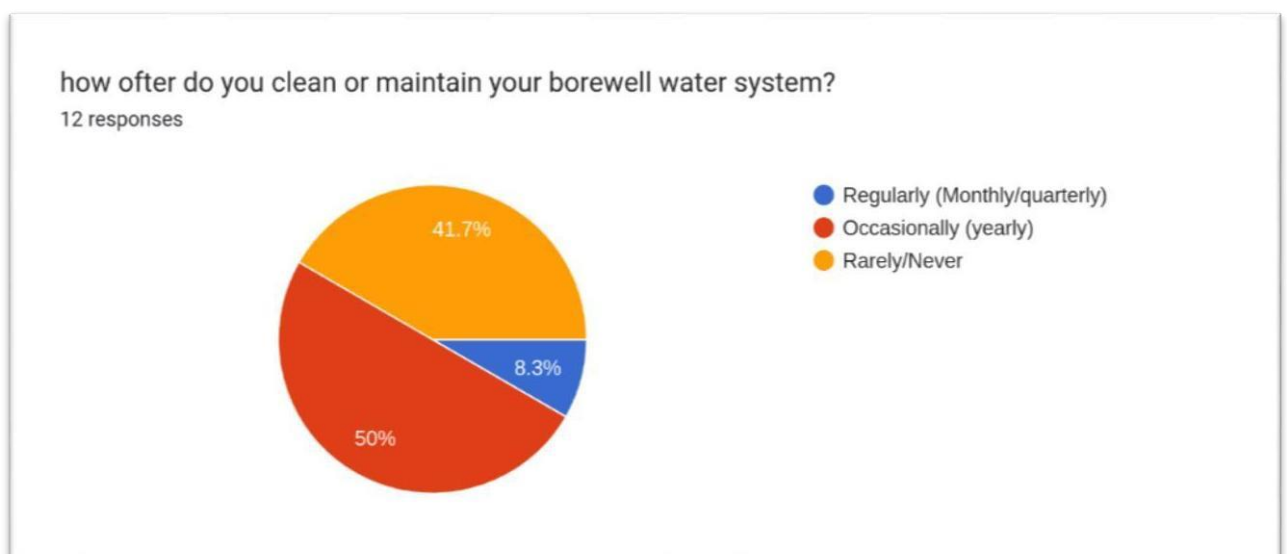


Fig 3.1.6 : Maintenance of Borewell Water System

The pie chart in Fig.3.1.6 indicates how often the respondents keep or clean up their borewell water system. 41.7% of 12 respondents keep it rarely or never, 50% keep it every now and then (once a year), and only 8.3% keep it regularly (monthly or quarterly).

Inference:

The questionnaire reveals an alarming trend of less frequent maintenance, since over 90% of individuals reported infrequently or rarely maintaining their borewell systems. Less frequent maintenance may result in the degradation of water quality and health risks. Promoting regular maintenance habits can significantly increase the reliability and safety of borewell water.

3.2 Literature Survey

Table 3.2.1 : Overview of Recent Research papers on IoT and Machine Learning for Water Quality Monitoring

Reference Number	Method	Dataset	Evaluation				Research Gaps
			Accuracy	Precision	Recall	F1-score	
[1]	ML (ANN, time-series)	USGS NWIS water data	NR	NR	NR	NR	Regression-based (R, RMSE); no classification
[2]	Physico-chemical analysis	Groundwater (Bangalore)	N/A	N/A	N/A	N/A	No ML or prediction
[3]	Hybrid IoT + ML using ANN & SVM	Lab-collected water samples (pH, DO, TDS, ORP, EC, turbidity, temperature)	ANN > SVM (exact values not reported)	ANN higher	ANN higher	ANN higher	Small dataset; no real-world data; limited ML models; limited contaminants
[4]	IoT + ML	Real-time sensor data	~95	~93	~92	~93	Energy efficiency, scalability
[5]	IoT + Deep Learning	Sensor + cloud data	96.8	95.9	96.1	96.0	High computation cost
[6]	IoT + ML	Smart monitoring data	NR	NR	NR	NR	Limited evaluation metrics
[7]	ML + geochemical indicators	Groundwater samples	~90	NR	NR	NR	No real-time IoT
[8]	IoT + Deep Learning	Sensor-based dataset	97.2	96.5	96.8	96.6	Limited geographic validation
[9]	IoT monitoring system	Real-time sensors	NR	NR	NR	NR	No ML prediction
[10]	ML models	Groundwater dataset	91.4	90.2	89.8	90.0	Offline analysis only
[11]	GIS + DRASTIC	Hydrogeological data	N/A	N/A	N/A	N/A	No ML / IoT
[12]	Physico-chemical analysis	Borewell samples	N/A	N/A	N/A	N/A	Traditional analysis
[13]	Review	Multiple studies	N/A	N/A	N/A	N/A	No experimentation

	paper						
[14]	Early IoT + ML	Water data	~85	NR	NR	NR	Outdated methods
[15]	IoT + Cloud	Sensor data	NR	NR	NR	NR	No ML evaluation
[16]	IoT + ML	Real-time water data	~88	NR	NR	NR	Small dataset
[17]	IoT + Cloud	Sensor dataset	NR	NR	NR	NR	Monitoring only
[18]	IoT + ML framework	Sensor + device data	92.3	91.6	91.1	91.3	Limited real-world deployment

The Table. 3.2.1 presents a comparative analysis of existing research on water quality monitoring and assessment using traditional methods, machine learning (ML), deep learning (DL), Internet of Things (IoT), and hybrid approaches. The comparison highlights key aspects such as methodology, dataset type, evaluation metrics, and identified research gaps. Earlier studies primarily relied on physico-chemical and statistical analyses without predictive capabilities, while recent works increasingly integrate IoT with ML and DL models to achieve higher accuracy, precision, recall, and F1-scores.

Comparative analysis



Fig 3.2.1 : Comparative Analysis Of Multiple Research Papers on Water Quality Analysis

The diagram in Fig. 3.2.1 presents a comparative overview of research approaches to water quality analysis with emphasis on three areas: Chemical Analysis, Machine Learning, and IoT-Based Monitoring.

- Chemical Analysis examines heavy metal contamination, physicochemical properties, microbiological factors and seasonality of water contents.
- Machine Learning Applications utilize prediction models, detailed data analysis and groundwater quality prediction to enhance water quality assessment.
- IoT-Based Monitoring integrates advanced neural networks, sensor-based data collection and real-time water monitoring using deep learning methods. This structured comparison indicates how traditional laboratory-based analysis is evolving towards sophisticated AI-based predictions and automated tracking networks, highlighting the increasing importance of technology in environmental studies.

3.3 Survey Outcome Summary

1. Participants:

- a. The survey was using the answers of various individuals (the demographics required by the survey were available in the answers).

2. Findings:

- a. **Participant's Rate of Awareness and Understanding:** Most of the respondents reported that they were aware/understood what was being surveyed, but there might be differences between that which was reported and what information was necessary.
- b. **Participant's Rate of Challenge:** The comment of the respondent was to reiterate the impediments to the present products and processes which are cost, availability and use of higher advanced technology.
- c. **Participant's Rate of Preference/Expectation:** A significant part of the respondents replied positively to the approaches and believed that these approaches would be cost effective and efficient as they focused more on automation and real-time monitoring as a major part of their expectation.

3. Implications:

- a. This survey has implications that point to the fact that innovative and readily accessible products and services are needed to deal with the aforementioned problems.
- b. Most critical to these users are environmental and affordability issues besides real time monitoring.

4. Actionable Insights:

- a. Certain portions of the information collected during this survey suggest that there is a market that the product solution could be developed to fill these existing gaps.
- b. Possibility of educational programme and demonstration to enhance awareness and knowledge on these issues.

4. PROBLEM FORMULATION

4.1. Problem Formulation

Background:

- Borewell water can be used as a major source of agriculture, domestic and industrial usage.
- Increased reliance on groundwater leads to the increased risks of contamination by heavy metals and nitrates and microbes.
- Pollution has severe health and environmental risks.

Issues with Existing Systems:

- The conventional techniques of detection such as drilling are invasive and expensive and damage the environment.
- Laboratory testing is time-consuming, labour intense and not real time.
- The existing systems cannot be used regularly or in large scale.
- There is a dire resource and accessibility problem in rural and semi-urban regions.

Need for Innovation:

- An efficient system of real-time, real and cost-effective nature is in dire need.
- The solution has to be scalable, environmental friendly and should have the ability to monitor continuously.
- There is an urgent need to have available real time and reliable data on the quality of borewell water to promote the health of people, agricultural stability and environmental safety.

4.2. Product Objectives

1. Real-time Monitoring:

Develop an automated, real-time monitoring system that sends out alerts whenever water quality or performance deviates from an established standard. Alerts can be used to quickly identify and report potential water contamination events so rapid response can be initiated.[15]

2. Accurate Detection:

To help agencies determine the sources and causes of potential contaminants, the latest technology in biosensing, the Internet of Things, as well as artificial intelligence (AI) will be used in the assessment and identification of multiple types of contaminants including heavy metals, microbes, nitrates and other chemicals of concern. All relevant data will be accessible via a single application.[17]

3. Predictive Analytics:

Implement AI and machine learning algorithms to analyze past contamination events and predict future contamination events; thus allowing for proactive responses to prevent both waterborne pathogens and water related crop loss.[3][17]

4. Cost-Effective Solution:

Design cost-effective, energy efficient, low-resource modular use systems that allow flexibility for growth and modification.[15]

5. Sustainability:

Limit the amount of material exposed to an environment by using only non-invasive, high-performance technologies that can be modified for operation in all types of geographies.

4.3. Applications of the Product

1. Agricultural Use:

Farmers in water, scarce areas will greatly benefit from a system that monitors the quality of borewell water used for irrigation by detecting the presence of harmful contaminants. Such a system will lead to healthier crops, higher quality yields, less soil degradation due to pollution, and ultimately the sustainability of freshwater resources [18].

2. Domestic Use:

Monitoring the quality of water extracted from borewells consistently for domestic use can give consumers a sense of trust that the water is safe for consumption at the minimum level. This will thus reduce the rate at which waterborne diseases are contracted, enhance the overall health of the public, and create a positive image of groundwater as a safe and reliable source of drinking water.

3. Industrial Use:

Borewell water constitutes a major element in the various manufacturing processes that occur in industries. Monitoring water quality is a good aspect that helps in meeting the standards set by various regulations, protects industries from damaging their equipment, as well as improves productivity.

4. Societal Impact:

This system improves access to safe and clean drinking water, addressing one of the major concerns for public health. It goes further to help in the realization of Sustainable Development Goals for water and sanitation-SDG 6-by facilitating the adoption of efficient and environmentally sustainable practices in managing the freshwater resources of the world [4].

4.4. Novelty

1. AI/ML Integration:

The trend analysis and prediction of the contamination events via the application of the newest Artificial Intelligence and Machine learning algorithms within the system to recognize the presence of unusual behavior of water quality parameters and make preventive measures during the process of reaching a better and faster decision.

2. Non-Invasive Techniques:

Determination of the amount with the quality of ground water through the application of the electric resistivity method and the GIS method without destroying the Environment. The non destructive technique is clean, fast and cost effective with regards to the destructive techniques applied in large scale.

3. IoT-Based System:

Integrating IoT based sensors to real-time data gathering and cloud based visualisation dashboard of the obtained data which enable the stakeholders to have practicable data based understanding and notifications through user friendly interfaces so that they can make timely decisions using the same.

4. Energy Efficiency:

Placement of energy-efficient sensor networks in such a way that they would have been able to work on a low power basis. The developed enhanced system architecture is steered towards reliable functioning in isolated and resources insufficient environments.

4.5. Scope of the Project

1. Domain Constraints:

- a. Other forms of water bodies like lakes, rivers or reservoirs will not be developed in the system.
- b. The main aim of the project is to aid real-time tracking of the water quality in the rural and semi-urban setups where facilities like sophisticated systems and techniques of carrying out such water quality check are barely accessible.

2. Application-Related Constraints:

- a. Technical challenges will arise during the successful execution of both software and equipment (IoT systems and ML), and the associated tasks will also need to be carefully designed to ensure that the elements integrate into each other to execute real-time monitoring.
- b. The distant places of deployment will pose logistical problems because of the inadequate infrastructure, access to resources, etc.

3. Extent of Implementation:

- a. The prototypes shall be created, which will be tested in the field tests to prove the system to be accurate, reliable, and efficient in real-life situations.
- b. The system is scalable since it is built in a modular way, hence, adaptable to deployment in large-scale in different geographic and/or socioeconomic contexts.

4. Future Prospects:

- a. The project could go beyond borewell water quality and could be also applied to other wider areas, e.g. real time monitoring of rivers, lakes and industrial water systems.

5. PROPOSED DESIGN

5.1 Proposed Model

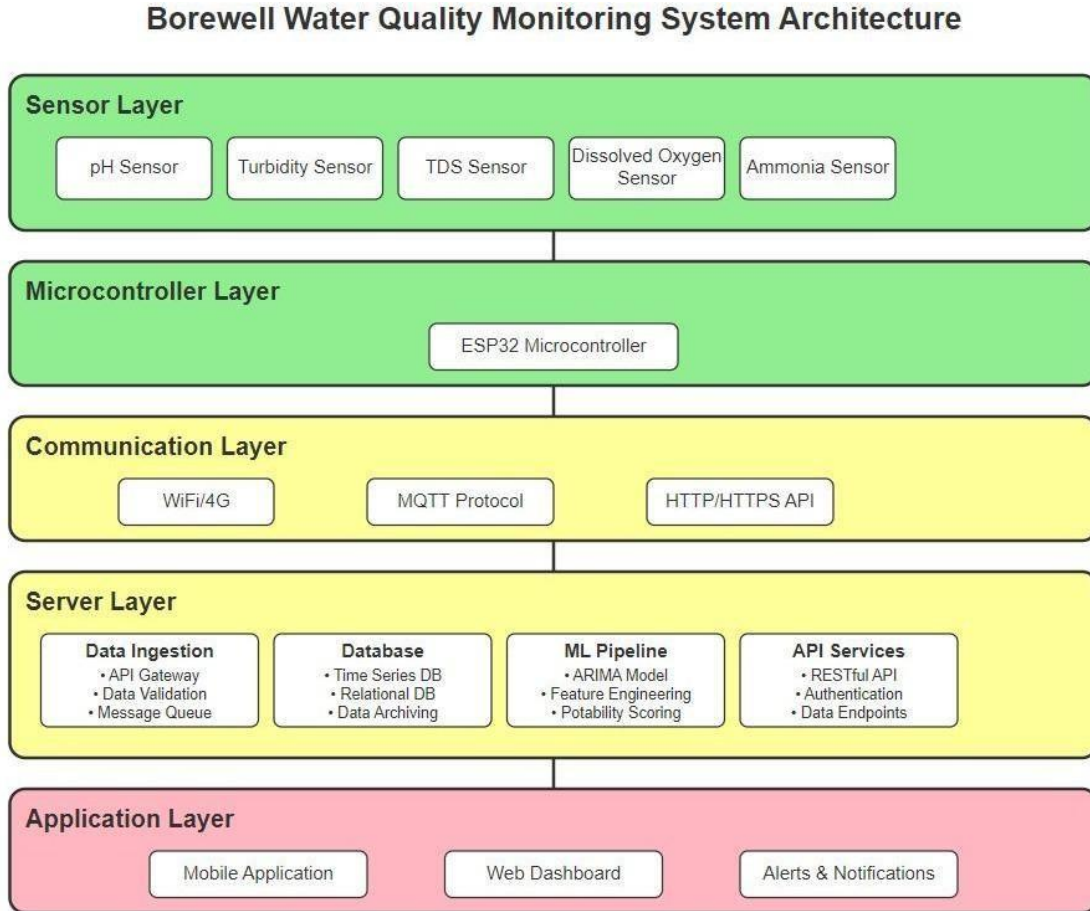


Fig 5.1 : Architecture of Borewell Water Quality Monitoring System

The Borewell Water Quality Monitoring System in Fig.5.1 has four layers sensor, communication, server and application. It uses sophisticated sensors to detect pH, turbidity, TDS, dissolved oxygen and ammonia to complete the analysis of water. MQTT and API support real-time data transmission, which allows improved connectivity. Cloud processing improves efficiency, and provides intelligent knowledge, traces of historical data, and automated warnings. The convenient mobile and web sites allow easy access to water quality control, generate reports, and make a well-informed decision regarding water safety and management.

Main Features

- **Sensors for Data Acquisition:**
 - **pH Sensor:** This is used to measure the acidity or the alkalinity of the water.
 - **TDS + EC Sensor:** Measures the total dissolved solids and electrical conductivity to determine purity of water.
 - **Turbidity Sensor:** This measures the suspended particles to identify the water clarity.
 - **Dissolved Oxygen (DO) Sensor:** Monitors oxygen concentration that is important in the evaluation of water health.
 - **MQ135 Sensor:** This sense measures the dangerous gases and volatile organic compounds (VOCs) used to identify the chemical pollution in water.
- **IoT Module:**
 - Getting real-time data provided by sensors.
 - Transmits information to a central server over wireless communication like Wi-Fi, LoRa or GSM.
- **AI/ML Integration:**
 - **Contamination Prediction:** Utilizes the past data to predict the water quality trends.
 - **Anomaly Detection:** Identifies abnormal behavior which may indicate contamination.
- **Real-Time Dashboard:**
 - **Visualization:** Displays parameters of water quality in user-friendly graphics and notifications.
 - **Alert System:** Alerts in case of any parameter going out of the safe limits.

General Functionalities

- **Information Collection:**
 - Water quality parameters are monitored continuously using the sensors and the data is stored.
- **Data Transmission:**
 - The IoT-enabled devices send data to the cloud or local servers for analysis.
- **Data Analysis:**
 - AI/ML algorithms process the data, forecast contamination and detect anomalies.
- **Real-Time Monitoring:**
 - Dashboards and mobile apps enable an easy view of timely water quality information.
- **Decision Support:**
 - Alerts and predictive insights assist users in taking preventive measures, such as water treatment or alternative sourcing.

5.2 Database Design

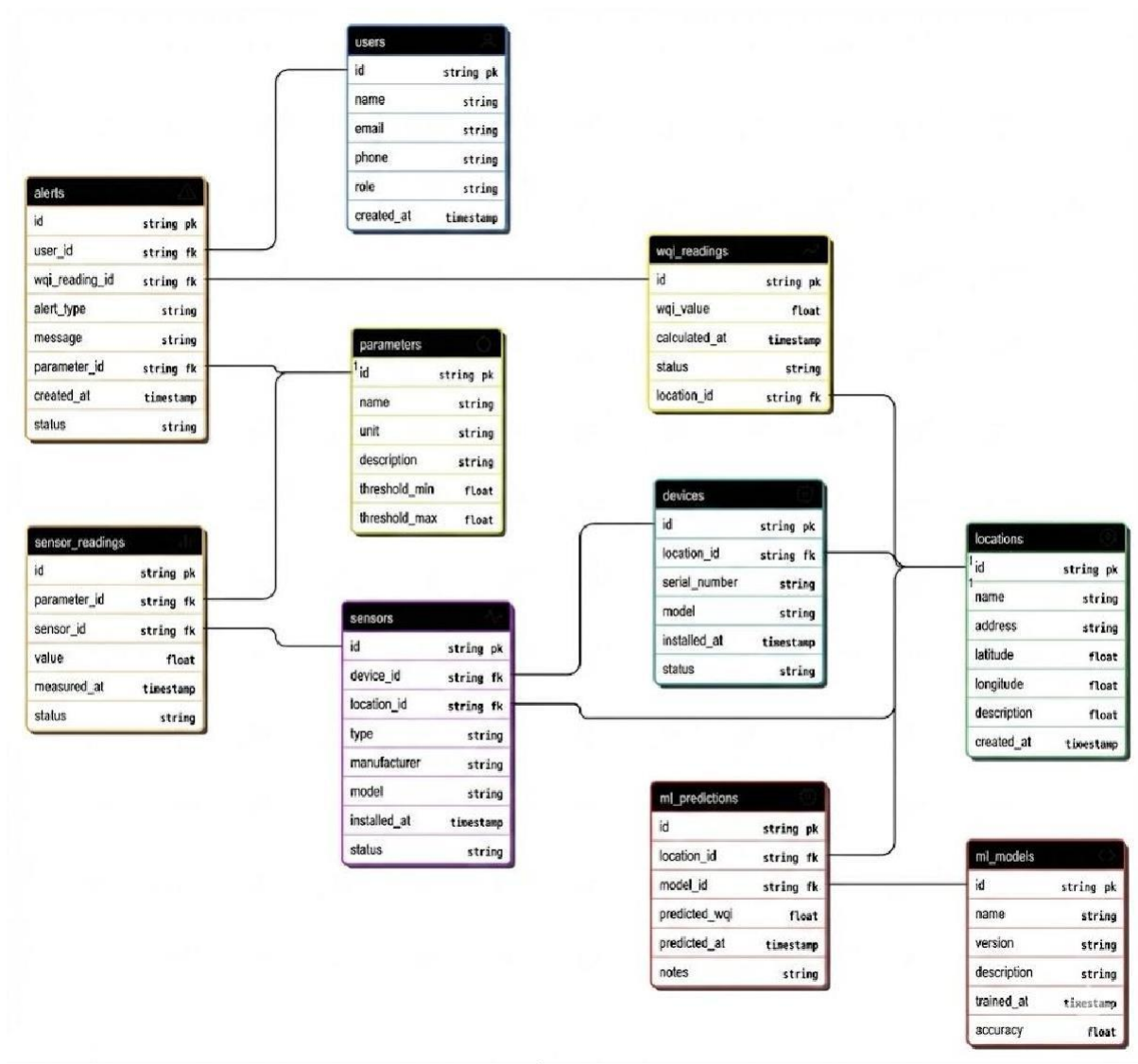
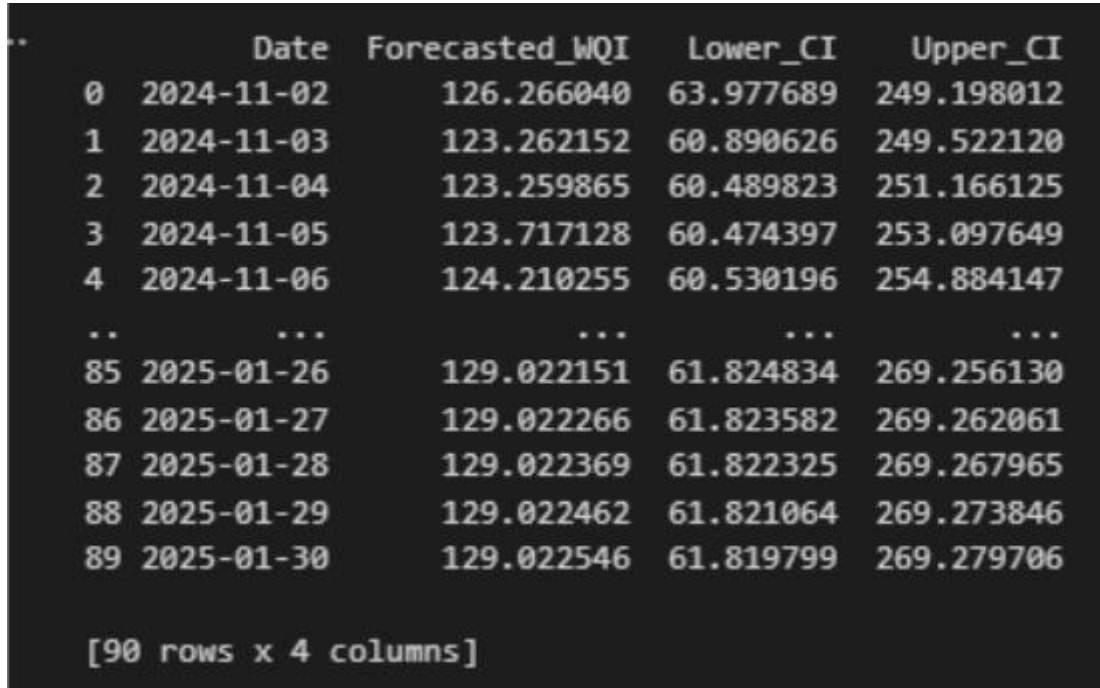


Fig 5.2 : Database Design

Fig. 5.2 presents a comprehensive database schema for an IoT-based monitoring and prediction system, illustrating how users, devices, sensors, data readings, alerts, and machine-learning components are logically connected.

6. IMPLEMENTATION

6.2. Modules implementation (Pseudocode / algorithm of the model)



	Date	Forecasted_WQI	Lower_CI	Upper_CI
0	2024-11-02	126.266040	63.977689	249.198012
1	2024-11-03	123.262152	60.890626	249.522120
2	2024-11-04	123.259865	60.489823	251.166125
3	2024-11-05	123.717128	60.474397	253.097649
4	2024-11-06	124.210255	60.530196	254.884147
..	...	...	...	...
85	2025-01-26	129.022151	61.824834	269.256130
86	2025-01-27	129.022266	61.823582	269.262061
87	2025-01-28	129.022369	61.822325	269.267965
88	2025-01-29	129.022462	61.821064	269.273846
89	2025-01-30	129.022546	61.819799	269.279706

[90 rows x 4 columns]

Fig 6.1 : Model Output

Fig. 6.1 shows the WQI values, ranging from approximately 123 to 129, predicted by the LSTM model over a time period of 3 months. The figure also displays the upper and the lower class intervals indicating the range of values in which the WQI occurs.

```

import pandas as pd
import numpy as np
from statsmodels.tsa.arima.model import ARIMA
import warnings
import pickle

# Suppress warnings
warnings.filterwarnings("ignore")

# Load dataset
df = pd.read_csv("cleaned_water_quality_with_date_and_wqi.csv")
df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values('Date')
df.set_index('Date', inplace=True)

# Time series of WQI
wqi_series = df['WQI']

# Log-transform the WQI series to ensure positive outputs
log_wqi_series = np.log(wqi_series)

# Fit ARIMA model on the log-transformed WQI
model = ARIMA(log_wqi_series, order=(2, 1, 2))
model_fit = model.fit()

# Forecast 90 future days
forecast_steps = 90
forecast = model_fit.get_forecast(steps=forecast_steps)

# Get predicted mean and confidence intervals (in log scale)
forecast_values_log = forecast.predicted_mean
confidence_intervals_log = forecast.conf_int()

```

Fig 6.2 : Model Code

Fig 6.2. Python code illustrating data preprocessing, log transformation of the WQI time series, and ARIMA(2,1,2) model fitting using Statsmodels library.

```

# Convert forecast and intervals back to normal scale using exp
forecast_values = np.exp(forecast_values_log)
lower_ci = np.exp(confidence_intervals_log.iloc[:, 0])
upper_ci = np.exp(confidence_intervals_log.iloc[:, 1])

# Generate future dates starting from the day after the last date
last_date = wqi_series.index[-1]
future_dates = pd.date_range(start=last_date + pd.Timedelta(days=1), periods=forecast_steps, freq='D')

# Combine forecast and intervals with dates
forecast_df = pd.DataFrame({
    'Date': future_dates,
    'Forecasted_WQI': forecast_values.values,
    'Lower_CI': lower_ci.values,
    'Upper_CI': upper_ci.values
})

# Save to CSV (optional)
forecast_df.to_csv("wqi_forecast_with_dates.csv", index=False)

# Show preview
print(forecast_df)

# Save the trained ARIMA model to a file
with open("arima_wqi_model.pkl", "wb") as f:
    pickle.dump(model_fit, f)

```

Fig-6.3 : Model Code

Fig 6.3. Python of WQI forecasting, inverse log transformation, confidence interval computation, future date generation, and export of results to a CSV file.

6.2.1 Introduction to WQI

The establishment of the Water Quality Index (WQI) provides a method for turning vast amounts of water quality data into one numerical value showing the general quality of that water. It allows individuals to quickly and clearly assess the safety of drinking water as well as the environmental impacts of using that water. The WQI is a useful tool in monitoring pollution patterns and comparing the water quality in different places and times.

6.2.2 Parameters Used

In this study, the WQI was calculated using the following key parameters extracted from the dataset:

- Dissolved Oxygen (DO)
- pH
- Electrical Conductivity
- Turbidity
- Ammonia (Ammonia-N)
- Hardness
- Sulphate
- Total Dissolved Solids (TDS)

These parameters are commonly monitored in water quality assessments and are indicative of the several pollution sources including industrial discharge, domestic wastewater and natural processes.

6.2.3 Methodology: Weighted Arithmetic Index Method

The WQI was computed using the weighted arithmetic index method, which involves the following steps:

6.2.3.1 Calculation of Quality Rating (Qi)

Each parameter is assigned a quality rating based on its deviation from the ideal value:

$$Q_i = \frac{(V_i - V_{\text{ideal}})}{(S_i - V_{\text{ideal}})} \times 100 \dots\dots\dots(1)$$

Where:

- V_i : Measured value of the parameter
- S_i : Standard permissible value (from WHO/BIS)
- V_{ideal} : Ideal value for pure water (e.g., 7 for pH, 0 for turbidity, etc.)

6.2.3.2 Calculation of Unit Weight (W_i)

Each parameter is assigned a weight inversely proportional to its standard permissible value:

$$W_i = \frac{K}{S_i} \quad \text{Where:}$$

$\frac{K}{S_i} W_i = K$ Where:

$$W_i = \frac{K}{S_i} \dots\dots\dots(2)$$

6.2.3.3 Final WQI

Calculation The WQI is computed as:

$$WQI = \frac{\sum W_i Q_i}{\sum W_i} \dots\dots\dots(3)$$

This single value provides a comprehensive assessment of water quality. Based on the WQI value as shown in Fig. 6.1 and the formula was used in code shown in Fig. 6.2 and 6.3, the water quality is categorized as follows:

Table 6.1 : Water Quality Indexes

WQI Range	Water Quality
0–25	Excellent
26–50	Good
51–75	Poor
76–100	Very Poor
>100	Unsuitable for Drinking

Table 6.1 describes the ranges of water quality index and the corresponding water quality for each range

Table 6.2 : Description of Formulas Used

Formula Name	Formula	Description
Quality Rating (Qi)	$Q_i = \frac{(V_i - V_{\text{ideal}})}{(S_i - V_{\text{ideal}})} \times 100$	Computes the quality rating for each parameter based on the measured value (Vi), standard permissible limit (Si), and ideal value for pure water (Videal).
Unit Weight (Wi)	$W_i = \frac{K}{S_i}$	Assigns a weight to each water quality parameter, inversely proportional to the standard permissible value (Si), with K as a constant.
Final WQI Calculation	$\text{WQI} = \frac{\sum W_i Q_i}{\sum W_i}$	Calculates the overall Water Quality Index (WQI) by summing the weighted quality ratings (WiQi) and dividing by the total sum of unit weights ($\sum W_i$).

Table 6.2 describes the methodology used to calculate the Water Quality Index (WQI). The quality rating (Qi) evaluates each parameter by comparing measured values with standard and ideal values, while the unit weight (Wi) assigns relative importance based on permissible limits. The final WQI is obtained by computing the weighted average of all quality ratings, providing an integrated assessment of overall water quality.

7. Experimentation And Results

7.1 Dataset / Tables (Description of fields of dataset)

Table 7.1 : Dataset of Parameters for Machine Learning Model

DO (mg/L)	pH	Conducti vity (mS/cm)	Turbidity (NTU)	Ammonia -N (mg/L)	Hardness (mg/L)	Sulphate (mg/L)	TDS (mg/L)
6.5	8.15	461	29	Less than 1	146	36	254
5.1	8.51	441	2.89	Less than 1	202	28	266
5	8.52	440	7.2	Less than 1	164	25	272
5	8.39	542	6.79	Less than 1	180	27	310
4.9	8.49	604	2	Less than 1	210	34	344
4.9	7.76	712	6.98	Less than 1	226	38	410
4.9	8.18	912	15.1	Less than 1	274	40	534
6.2	8.44	420	6	Less than 1	146	15	247
6	8.3	468	9	Less than 1	155	12	284
6	8.34	265	7	Less than 1	105	8	159
5.8	8.28	295	2	Less than 1	128	10	185
4.9	8.11	342	4	1	125	12	212
5.2	8.41	437	2.2	Less than 1	142	22	244
4.8	7.68	289	5	Less than 1	120	5	176

The dataset in Table. 7.1 has been referenced from <https://data.telangana.gov.in/node/202046> and appropriately modified to suit the needs of our project

TDS +EC Sensor with Arduino UNO



Fig 7.1 : Total Dissolved Substances and Electrical Conductivity Sensor

Description :

The TDS sensor (Total Dissolved Solids), as shown in Fig. 7.1, measures how much stuff is dissolved in water. TDS is measured in ppm (parts per million). Low TDS = Very pure. Medium TDS = Typical drinking water. High TDS (>1000 ppm) = Potentially not safe to drink.

The EC (Electrical Conductivity) sensor, as shown in Fig. 671, measures the ability of water to conduct electricity, indicating the presence of dissolved ions like salts and minerals. It is typically measured in $\mu\text{S}/\text{cm}$ or mS/cm , where higher EC values suggest water with more dissolved solids, often indicating harder or saltier water as shown in Fig 7.2.

Output :

```
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:0ppm
13:57:10.462 -> TDS Value:3ppm
13:57:10.462 -> TDS Value:5ppm
13:57:10.462 -> TDS Value:5ppm
13:57:10.462 -> TDS Value:8ppm
```

Fig 7.2 : Output of the TDS and EC Sensor

Turbidity Sensor with Arduino UNO

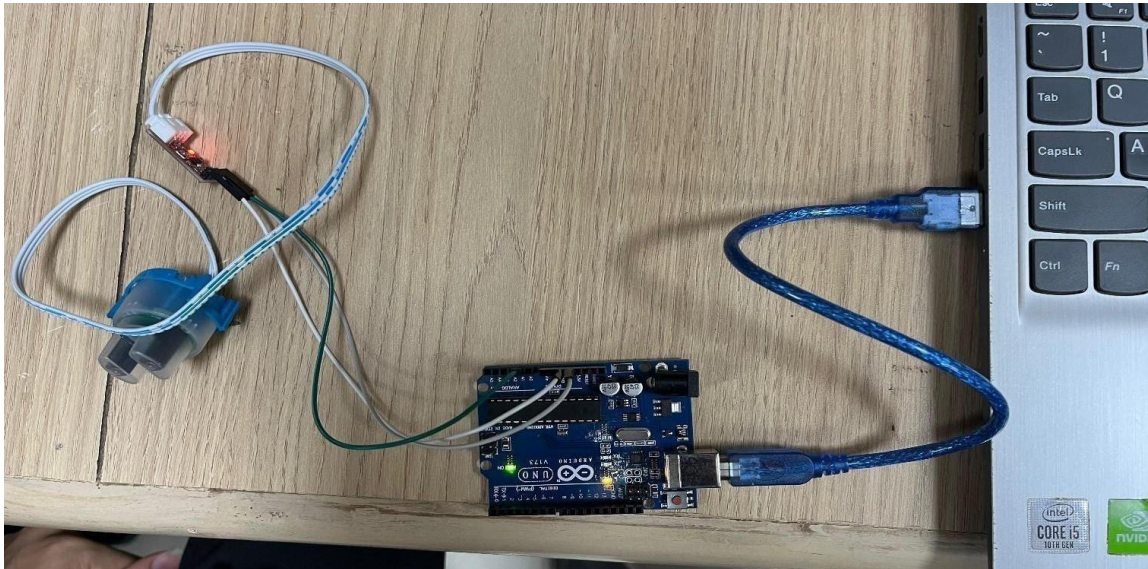


Fig 7.3 : Turbidity Sensor

Description:

A turbidity sensor, as shown in Fig. 7.3, is used to measure how clear or dirty water is.

It works by sending light through water and checking how much light comes back more dirty

= less light passes. How it's used with Arduino: It gives an analog voltage output. Clean water → Higher voltage. Dirty water → Lower voltage.

Output :

```
13:24:47.496 -> raw:      58      Voltage (mv) 283
13:24:48.515 -> raw:      86      Voltage (mv) 419
13:24:49.491 -> raw:      89      Voltage (mv) 434
13:24:50.511 -> raw:      89      Voltage (mv) 434
13:24:51.529 -> raw:      89      Voltage (mv) 434
13:24:52.507 -> raw:      89      Voltage (mv) 434
13:24:53.522 -> raw:      90      Voltage (mv) 439
13:24:54.495 -> raw:      84      Voltage (mv) 410
13:24:55.524 -> raw:      70      Voltage (mv) 341
13:24:56.548 -> raw:     744      Voltage (mv) 3632
13:24:57.522 -> raw:     747      Voltage (mv) 3647
13:24:58.545 -> raw:     748      Voltage (mv) 3652
13:24:59.517 -> raw:     740      Voltage (mv) 3613
13:25:00.537 -> raw:     738      Voltage (mv) 3603
13:25:01.555 -> raw:     721      Voltage (mv) 3520
13:25:02.528 -> raw:       7
```

Fig 7.4 : Output of the Turbidity Sensor

MQ-135 Gas Sensor (Ammonia based Sensor) with Arduino UNO

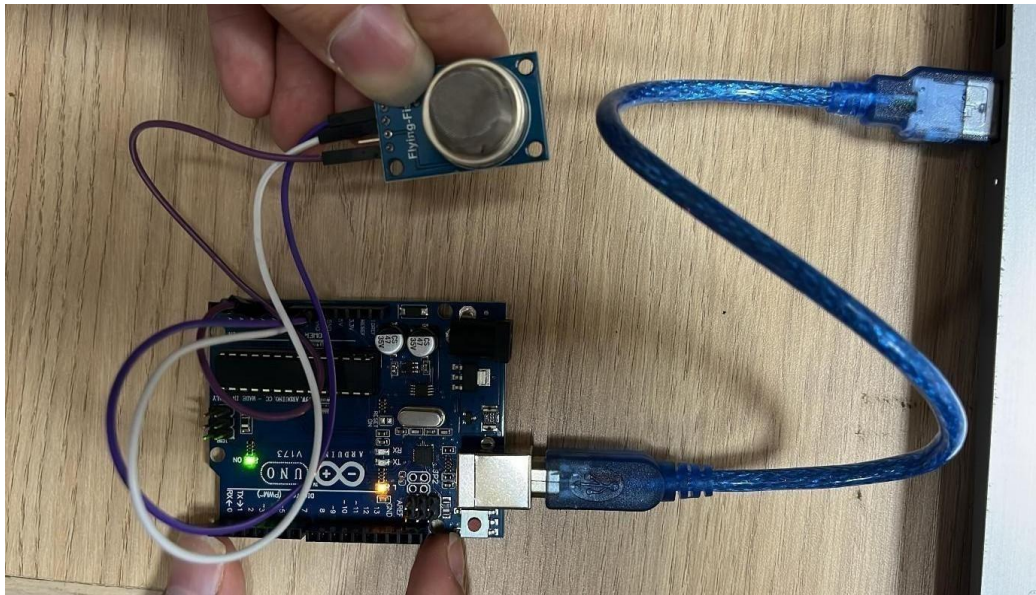


Fig 7.5 : MQ-135 Ammonia gas sensor

Description :

As shown in Fig. 7.5, The MQ-135 is a gas sensor that measures air quality using Arduino. The MQ-135 can recognise gases such as Ammonia, Nitrogen oxides, Alcohol, Benzene, Smoke and CO₂. Basic operation with Arduino – the MQ-135's analog voltage output is proportional to the concentration of each of these gases. You connect the MQ-135's output pin to one of the Arduino's analog pins..

Output :

```

49 | delay(500);
50 | }
51 |

```

Output Serial Monitor X

Message (Enter to send message to 'ESP3253 Dev Module' on 'COM5')

Raw Value: 166	Δ from baseline: -84
Raw Value: 159	Δ from baseline: -91
Raw Value: 151	Δ from baseline: -99
Raw Value: 116	Δ from baseline: -134
Raw Value: 71	Δ from baseline: -179
Raw Value: 70	Δ from baseline: -180
Raw Value: 175	Δ from baseline: -75
Raw Value: 251	Δ from baseline: 1
Raw Value: 342	Δ from baseline: 92
Raw Value: 436	Δ from baseline: 186

Fig 7.6 : Output of MQ-135 sensor

8. CONCLUSION

This project effectively points out how critical it is to have dependable, real, time, and environmentally friendly borewell water quality monitoring in areas that rely on groundwater for their needs. Through a combination of IoT, based sensing, AI/ML, enhanced predictive analytics, and non, invasive survey methods, the system proposed in this paper essentially overturns the main drawbacks of conventional water testingsuch as high cost, time, consuming results, manual sampling, and lack of continuous monitoring.

By leveraging advanced sensors, cloud connectivity, and analytical models, not only accurate measurement of water quality parameters is ensured but also their intelligent interpretation through which contamination trends can be forecasted and the release of early warnings can be prompted. This, in turn, results in an increase of the level of public health, the support of agricultural productivity, and the facilitation of industries in maintaining their regulatory compliance.

Besides that, the integration of renewable energy solutions such as solar power makes the system environmentally sustainable in the long run and at the same time economically viable in the case of remote and resource, poor areas.

All in all, the project lays down a solid groundwork for the revolution of groundwater monitoring from an interval, sampling task to a continuous, data, driven, and community, empowered one. The solution pledges to be consistent with the worldwide sustainable development agenda and it serves as an example of how technological advancements can be instrumental in securing water as an input and a living resource for society in the long run.

9. REFERENCES

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10. APPENDIX

- 1. Quality Analysis of Borewell Water ,Research Paper presented at ICATES '25 Conference , St. Xavier's College Mahim.**
- 2. Participated at Shark Tank Fr. CRCE 2025.**

